FABIAN ALVAREZ-PRIMO, PhD

MATERIALS ENGINEER & PRODUCTION LEADER | METALLURGY & FAILURE ANALYSIS | HIGH-RATE OPERATIONS UNDER PRESSURE

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Summary

Metallurgical and materials engineer (B.S., Ph.D., UTEP) with deep expertise in failure analysis, materials characterization, and manufacturing hardware, who has deliberately built a second core strength: leading people under pressure. Took on Amazon operations to lead large teams where quality, safety and deadlines don't give. Now leads a **60+ associate** high-rate operation at **99.7% quality compliance**, a **0.0% serious incident rate** for 9+ months, and an **82% throughput increase**, still engineering custom Python/SQL/JS tooling to find the bottleneck. Brings both to Starship: reads drawings, fracture surfaces, and data, and leads the floor that builds the hardware.

Technical Skills

- **Metallurgy & materials:** physical metallurgy and alloys, metals processing, hands-on metal casting, corrosion, phase transformations, mechanical behavior of materials
- **Failure analysis & characterization:** SEM fractography, EDS, XRD, XRF, TEM, Raman/FTIR; root cause investigation; microstructure analysis
- **Testing:** tensile, compression, and creep testing to ASTM; TGA, DMA; density, weight-loss, and compressive-strength testing; ASTM test-method development
- **Manufacturing & design:** additive manufacturing and 3D printer hardware, injection molding, extrusion; SolidWorks, Fusion 360, AutoCAD; FEA (FEniCS); SOPs and work instructions
- **Production systems:** lean and standard work, root cause and corrective action, labor and capacity planning, KPI design, quality control (CLIA/COLA, cGMP, ISO 9001)
- **Software & data:** Python (pandas, scikit-learn, Streamlit), SQL, R, MATLAB, JavaScript (browser automation), Excel / Google Sheets (advanced), Tableau

Professional Experience

Operations Manager II (Area Manager, L5) | Amazon | Savannah, GA | 2024 – Present

Chose high-volume operations to lead large teams in a high-stakes, high-pressure environment: hard hourly departure deadlines, heavy equipment, and zero tolerance on safety.

- Lead a **60+ associate** operation across a compressed 4-day schedule through Process Assistants and floor leads, holding **99.7% quality compliance** against hard departure schedules.
- Raised throughput **82% (220 → 400+ units/labor hr)** and cut recirculation from **5% to below 2%**. Treated the dock as an engineering system: instrumented it, found the station-level constraints, and redesigned staffing and material paths around them.
- Engineered a three-layer flow-visibility system (browser-automation scraper of live WIP data, a real-time analytics layer for root causes, staffing ratios, DPMO and hourly throughput, and a staffing/rate tool), replacing manual workflows.
- Built Python labor-forecasting models and a statistical performance tool (curve-fit throughput by process path) used in production for ~8 months to plan headcount and overtime and to remove subjectivity from reviews.
- Sustained a **0.0% Serious Incident Rate for 9+ consecutive months**, and beat the DPMO goal by **29% (5,291 vs. 7,500)** by tracing defects to root cause and changing the process.
- Stood up Vendor Returns operations from scratch: wrote the process, set the standard work, and trained the team.
- Developed **15 Process Assistants** (up to **9 concurrently**) and **8 new leaders** into full role ownership. Ran Ship Dock and the DEA team while the Operations Manager was absent, and was selected to lead a best-practice share-out for peer Area Managers on stand-ups, real-time feedback, and building trust.

Account Executive | DISH Network | 2023 – 2024

Chose a quota-driven, customer-facing role to build communication, persuasion, and composure under pressure.

- Won a company-wide award for the highest contracts sold during training; promoted to Account Executive.
- Built Tableau analytics to drive customer-retention strategy.

Post-doctoral Research Fellow | Texas Tech University Health Sciences Center | El Paso, TX | 2022 – 2022

- Ran postmortem human brain biobanking operations and clinical research in neuropsychiatry with the Southwest Brain Bank and El Paso County Coroner's Office, covering specimen intake, processing, controlled storage, and chain-of-custody documentation.
- Led a stem-cell reprogramming project under **cGMP-compliant** protocols and verified the integrity of stored tissue through follow-up culture testing.
- Owned statistical analysis (R, Python) across concurrent studies and coordinated deliverables across clinical, molecular, and neuroscience teams; co-authored *Frontiers in Psychiatry* (2023).

Clinical Laboratory Scientist & Supervisor | West Texas Pain Institute | El Paso, TX | 2021 – 2022

- Supervised a team of **9** that scaled sample processing **150%**, averaging **1,300+ patient samples per month** under full **CLIA and COLA** compliance, where a documentation error is a patient-safety issue.
- Operated and maintained LC/MS, immunoassay, and DNA sequencing platforms; enforced OSHA protocols and controlled documentation.

Doctoral Research Assistant | Inspired Materials & Stem Cell Tissue Engineering Lab, UTEP | El Paso, TX | 2017 – 2020

- Improved 3D printer manufacturing precision **30%** through hands-on hardware modifications.
- Co-authored a failure analysis study of additively manufactured specimens (*Journal of Failure Analysis and Prevention*, 2019); developed and characterized a conductive composite to ASTM and **ISO 9001** standards.
- Ran lab operations across **5+ concurrent projects** and trained **30+ students** on SEM and 3D printing.

R&D Engineer Intern | Climax Molybdenum | Sahuarita, AZ | 2015 – 2015

- Developed ASTM test methods for a refractory ceramic in leaching-vessel service (TGA, density, weight-loss, compressive strength), led an initiative to extend its service life, and wrote the material-handling SOP.

Undergraduate Research Assistant | W.M. Keck Center for 3D Innovation, UTEP | 2014 – 2015

- Tested tensile, compression, and creep properties of aerospace-grade ABS/SEBS composites with the U.S. Air Force and America Makes; characterized fracture surfaces with SEM and XRD on ASTM D638 specimens. Contributed to the *Wohlers Report 2016* (Material Properties appendix).

Engineering Projects

- **Imaging-to-CAD pipeline:** DICOM → STL scaffold-geometry pipeline, ~3,000 lines of Python with a 789-line test suite and ISO 14971 risk documentation.

Education

Ph.D., Materials Science & Engineering | The University of Texas at El Paso | 2020 | GPA 3.91

B.S., Metallurgical & Materials Engineering | The University of Texas at El Paso | 2015

Publications & Honors

7 peer-reviewed publications, including failure analysis and materials characterization. Tau Beta Pi · Alpha Sigma Mu · Alpha Chi · Society for Biomaterials (President, 2 terms)